

Storage Devices -- Guided Notes ANSWER KEY

Unit 2.3 | CompTIA Network+ N10-009 Obj. Core 1 3.4 | N10-008

For instructor use / distribute after students complete the notes.

Question 1

A 5400 RPM HDD is typically used in _____ drives due to lower heat and power consumption. A 7200 RPM HDD provides _____ performance and is common in _____ storage applications. S.M.A.R.T. stands for _____ and is a built-in system that monitors _____ health and predicts potential failure.

Answer: Laptop (mobile); better read/write; desktop; Self-Monitoring, Analysis, and Reporting Technology; drive.

Question 2

A _____ sound coming from a hard drive (HDD) is a symptom of mechanical failure -- a read/write head is likely contacting the _____. The correct immediate response is to _____ all important data before the drive fails completely.

Answer: Clicking (grinding); platters (disk surface); back up (copy / save).

Question 3

SATA III has a maximum transfer speed of approximately _____ MB/s. An NVMe SSD using PCIe 4.0 can reach speeds of up to _____ MB/s. This means NVMe is roughly _____ times faster than the fastest SATA SSD for sequential reads.

Answer: 550 MB/s (SATA III); 7000 MB/s (PCIe 4.0 NVMe); approximately 12-13 times faster.

Question 4

An M.2 SSD with the designation 2280 is _____ mm wide and _____ mm long. The M.2 form factor connects directly to the motherboard using an _____ slot, eliminating the need for a _____ cable. M.2 drives can use either the SATA or _____ protocol depending on the type.

Answer: 22 mm wide; 80 mm long; M.2; SATA data cable; NVMe.

Question 5

RAID 0 stripes data across _____ or more drives for improved _____. It provides _____ (does/does not) redundancy. RAID 1 mirrors data across _____ drives, providing _____ redundancy. If one drive fails in RAID 1, the data is _____ from the mirror.

Answer: Two; performance (speed); does NOT provide redundancy. RAID 1: two drives; full (complete); recovered.

Question 6

RAID 5 requires a minimum of _____ drives and distributes _____ data across all drives. If one drive fails, data can be reconstructed using the _____ information. RAID 5 provides both performance and redundancy but _____ (can/cannot) survive a two-drive failure.

Answer: Three (3); parity; parity; cannot survive a two-drive failure.

Question 7

FAT32 has a maximum individual file size of _____ GB. This means you _____ (can/cannot) store a 30 GB video file on a FAT32-formatted drive. NTFS and exFAT do not have this limitation. GPT partition style is required when the drive is larger than _____ TB and supports up to _____ primary partitions.

Answer: 4 GB; cannot; 2 TB; 128 primary partitions.

Question 8 -- Real World Application

REAL WORLD: A small business owner asks you to set up storage for a file server that needs to survive a single drive failure, maximize storage efficiency, and hold files larger than 4 GB. They have four identical 4 TB drives available. Recommend a RAID level, explain your reasoning, calculate the usable storage, and specify the required file system.

Answer: Recommend RAID 5. It uses three or more drives, distributes parity across all drives, survives one drive failure, and provides better storage efficiency than RAID 1. With four 4 TB drives in RAID 5: usable storage = $(4 - 1) \times 4 \text{ TB} = 12 \text{ TB}$ (one drive worth used for parity). RAID 1 would yield only 8 TB usable (50% efficiency vs. 75% for RAID 5). File system: NTFS -- required for files over 4 GB (FAT32 4 GB limit) and for Windows file server permissions. Note: RAID is not a backup -- a separate offsite backup should still be maintained.